

SOLID MECHANICS FOR AERONAUTICS LAB

III Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE06	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100

LIST OF EXPERIMENTS

1. Direct Tension Test on UTM
2. Deflection test on Simple supported Beam
3. Deflection test on Cantilever Beam
4. Torsion Test
5. Brinell hardness test
6. Compression test on cube
7. Test on springs
8. Impact Strength-Izod Test
9. Impact Strength-Charpy Test
10. Shear Test on UTM

Reference Books:

1. R. K. Bansal (2011), A Textbook of Fluid Mechanics and Hydraulic Machines, 10th edition, Laxmi Publications, New Delhi, India.
2. Ramamrutham. S (2012), Strength of materials, 17th edition, Dhanpat Rai Publications,

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Analyze the behavior of the solid bodies subjected to tensile loading on the axial member and determine the stress, strain and modulus of elasticity.
2. Apply knowledge of materials and structural elements to the analysis of simple structures of beam subjected to transverse loading and determine deflection on the beam
3. Analyze the knowledge of the shaft subjected to torsion loading on the circular shaft and determine the shear stress and modulus of rigidity.
4. Analyze the behavior of the coil spring and determine the stiffness of a compression and tension spring.
5. Analyze the behavior of the material undergoing various types of shear test, impact test and compression test and determine the failure strength of the material.